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Emerging Pathogens in Planetary Health and Lessons from Comparative Genome Analyses of Three *Clostridia* Species

Ankit Singh Tanwar,^{1,2,*} Padival Shruptha,^{3,*} Apoorva Jnana,⁴ Angela Brand,^{1,2,5} Mamatha Ballal,⁶
Kapaettu Satyamoorthy,³ and Thokur Sreepathy Murali⁴

Abstract

Clostridioides difficile (CD) is a major planetary health burden. A Gram-positive opportunistic pathogen, CD, colonizes the large intestine and is implicated in sepsis, pseudomembranous colitis, and colorectal cancer. *C. difficile* infection typically following antibiotic exposure results in dysbiosis of the gut microbiome, and is one of the leading causes of diarrhea in the elderly population. While several studies have focused on the toxigenic strains of CD, gut commensals such as *Clostridium butyricum* (CB) and *Clostridium tertium* (CT) could harbor toxin/virulence genes, and thus pose a threat to human health. In this study, we sequenced and characterized three isolates, namely, CT (MALS001), CB (MALS002), and CD (MALS003) for their antimicrobial, cytotoxic, antiproliferative, genomic, and proteomic profiles. Although *in vitro* cytotoxic and antiproliferative potential were observed predominantly in CD MALS003, genome analysis revealed pathogenic potential of CB MALS002 and CT MALS001. Pangenome analysis revealed the presence of several accessory genes typically involved in fitness, virulence, and resistance characteristics in the core genomes of sequenced strains. The presence of an array of virulence and antimicrobial resistance genes in CB MALS002 and CT MALS001 suggests their potential role as emerging pathogens with significant impact on planetary health.

Keywords: planetary health, whole genome sequencing, omics, antimicrobial resistance, virulence, pangenome analysis

Introduction

CLOSTRIDIODES DIFFICILE (CD) is a Gram-positive opportunistic pathogen that resides in the human gut. It causes severe gastrointestinal diseases such as diarrhea, pseudomembranous colitis, as well as sepsis and multiple organ dysfunction with major outbreaks worldwide, and is the most significant cause of diarrhea in the elderly population (Jin et al., 2017; Leffler and Lamont, 2015). Treatment

with broad spectrum-antibiotics and longer periods of hospitalization result in dysbiosis of normal gastrointestinal flora and creates a niche suitable for overgrowth of CD (Rineh et al., 2014). Continued use of broad-spectrum antibiotics and old age are major risk factors for *C. difficile* infections (CDI).

Reports suggest at least 1–3% of hospitalized patients acquire CDI, while 25% of these patients experience recurrent infections due to high antibiotic resistance among these strains. Approximately half-a-million CDI cases are reported

¹Department of Public Health Genomics, Manipal School of Life Sciences, Manipal Academy of Higher Education, Manipal, India.

²United Nations University–Maastricht Economic and Social Research Institute on Innovation and Technology (UNU-MERIT), Maastricht, The Netherlands.

³Department of Cell and Molecular Biology, Manipal School of Life Sciences, Manipal Academy of Higher Education, Manipal, India.

⁴Department of Biotechnology, Manipal School of Life Sciences, Manipal Academy of Higher Education, Manipal, India.

⁵Faculty of Health, Medicine and Life Sciences, Maastricht University, Maastricht, The Netherlands.

⁶Department of Microbiology, Kasturba Medical College, Manipal Academy of Higher Education, Manipal, India.

*Equal first authors.

every year from the United States alone, with health care costs for treatment approaching US\$1.5 billion (Chen et al., 2017), while all-cause mortality within a month of infection is reported to be 20% (Smits et al., 2016). However, most individuals with CD remain asymptomatic and only few exhibit severe disease symptoms. CD spores are highly resistant and upon exposure to bile salts in the gastrointestinal tract, germinate to form vegetative cells, which colonize the colon (Edwards et al., 2016).

In clinical settings, CDI is generally diagnosed by culture-based methods that involve fecal material, testing for non-specific antigens such as glutamate dehydrogenase and/or specific toxins (toxins A and B) by cytotoxicity or immunoassays, which are time consuming and technically demanding, and show low sensitivity and specificity (Bartlett and Gerding, 2008). The gold standard assays take longer time to diagnose the infection, while severe cases might require immediate surgical intervention. Molecular techniques such as polymerase chain reaction (to detect toxin genes such as *tcdA* and *tcdB*) (Chen et al., 2017), 16S rRNA ribotyping (Bidet et al., 1999), pulsed-field gel electrophoresis, and microarray analysis (Al-Khalidi et al., 2012) have been used for the diagnosis and delineating different species of *Clostridioides*. Molecular assays provide a distinctive advantage over culture-based methods with their rapid turn-around time, in addition to identifying asymptomatic carriers often missed by culture assays.

However, exclusive reliance on molecular tests without detecting the presence of toxins can lead to false positives (Smits et al., 2016). Asymptomatic carriage has also been reported, which further complicates detection of pathogenic *Clostridia* (Cassir et al., 2016a). The presence of asymptomatic carriers in the hospital settings is one of the major reasons for epidemic outbreaks and the nosocomial nature of CDI (Riggs et al., 2007). In addition, the severity of CDI cannot be explained merely by the genetic factors of CD alone. CDI is often associated with dysbiosis of gut microbiome, which exponentially worsens the infection (Abbas and Zackular, 2020). Hence, it is critical to study the interactions of CD with co-occurring pathogens such as *Clostridium butyricum* (CB) and *Clostridium tertium* (CT) (Ferraris et al., 2012).

In addition, study of the genomes of co-occurring pathogens may offer insights into pathogen evolution and improve our understanding of the genetic basis of various processes critical for pathogenesis such as virulence and resistance. It will be interesting to explore genomes of CB, a probiotic producing human gut commensal, with reports of being both beneficial and harmful (Cassir et al., 2016b; Kanai et al., 2015), and CT, a largely nontoxigenic strain occasionally associated with cases of bacteremia, meningitis, and several others (Kourtis et al., 1997; Moore and Lacey, 2019; Shah et al., 2016) for pathogenic elements. This can provide insights into the emergence of pathogens with improved virulence, antimicrobial resistance (AMR), and overall pathogenic potential.

The goal of this work was to study the genomes of CT (MALS001) and CB (MALS002) and compare it with that of CD (MALS003) for genomic surveillance to uncover virulence-associated factors and their potential role as emerging pathogens. Emerging pathogens have the capacity to spread, infect large populations, devastate health systems,

raise morbidity and mortality rates, and add to the economic burden, making them a serious planetary health concern.

Materials and Methods

This study was approved by the Kasturba Medical College and Kasturba Hospital Institutional Ethics committee in Manipal, India. Informed consent was obtained from all individuals included in the study.

Culturing of *Clostridia* isolates

For this study, a clinical isolate of CT (MALS001) and CB (MALS002) was isolated from stool samples of patients visiting Kasturba Hospital, Manipal (Enteric Diseases Division, Department of Microbiology, Kasturba Medical College, Manipal) and CD MALS003 (from CD ATCC 9689) was included. The strains were revived in Brain Heart Infusion agar (BHIA) (HiMedia, Mumbai, India) and incubated anaerobically at 37°C in a GasPak EZ Anaerobe system (Becton Dickinson, USA) for 48 h.

Starch hydrolysis assay

Cultures of isolates CB MALS002, CD MALS003, and CT MALS001 ($OD_{600} - 0.1$) were streaked onto BHIA plates supplemented with 2% starch (HiMedia) and incubated at 37°C for 16 h. Culture plates were flooded with Gram's iodine solution and examined for the presence of clear zones (Mahon and Manuselis, 1995).

Motility test

The single colonies of CB MALS002, CD MALS003, and CT MALS001 grown on BHIA were picked using straight wire and stab inoculated into 0.175% BHIA in a glass vial. Following incubation at 37°C for 16 h, vials were removed from the anaerobic chamber and photographed to record the swimming motility (Chen et al., 2019; Dingle et al., 2011).

Minimal inhibitory concentration assay

Minimal inhibitory concentration (MIC) testing was performed using the gradient method, wherein plastic strips with predefined concentrations of antibiotic are employed to determine the exact MIC for chosen antibiotics (Kowalska-Krochmal and Dudeck-Wicher, 2021). In this study, MIC assay was done for ciprofloxacin and vancomycin based on the corresponding presence of resistance genes in the genomes of isolates sequenced in this study. Toward this, following revival of CB MALS002, CD MALS003, and CT MALS001 as described before, isolates were streaked ($OD_{600} - 0.1$) onto Mueller Hinton Agar plates. HiComb gradient MIC strips (HiMedia) of vancomycin (0.016–256 µg gradient) and ciprofloxacin (0.001–240 µg gradient) were placed at the center of the inoculated plates, which were subsequently incubated at 37°C in a GasPak EZ Anaerobe system (Becton Dickinson) for 48 h. Average MIC values obtained from triplicates for each strain were assessed for AMR.

Interpretation of AMR was performed as per the Clinical and Laboratory Standards Institute (CLSI) and the European Committee on Antimicrobial Susceptibility Testing (EUCAST) guidelines, where applicable. For ciprofloxacin, interpretation was done with breakpoints assigned for

moxifloxacin defined by CLSI (2020, 30th edition) since neither CLSI nor EUCAST had interpretation criteria for ciprofloxacin (Büchler et al., 2014). MIC values of ≤ 2 , 4, and ≥ 8 μg of ciprofloxacin indicate sensitivity, intermediate resistance, and resistance accordingly (CLSI guidelines for moxifloxacin were considered). For vancomycin, MIC values of ≤ 2 and > 2 μg indicate sensitivity and resistance accordingly (Büchler et al., 2014; EUCAST, 2023).

Biofilm formation analysis

Biofilm formation is a key ability that enhances the pathogenesis of CD (Vuotto et al., 2018). CB MALS002, CD MALS003, and CT MALS001 strains were assessed for biofilm formations in anaerobic static condition at 24 and 48 h postincubation. Tissue culture plate method was used to determine the static biofilm by staining with 0.1% crystal violet.

Interspecies interactions have been reported to enhance CD virulence and resistance (Abbas and Zackular, 2020). Hence, polymicrobial biofilms [pair-wise combinations of the three isolates (CB+CD, CT+CD, and CB+CT) and multiple combinations (CB+CD+CT)] were also assessed by tissue culture method. Briefly, following culturing as before in Brain Heart Infusion broth, 100 μL of each isolate was combined with the other at an OD₆₀₀ of 0.05 for pair-wise and multiple interactions. Two hundred microliters of each combination was seeded into a 96-well plate. Following seeding, plates were incubated for 24 and 48 h followed by staining with 0.1% crystal violet (Prasad et al., 2020).

Cytotoxicity assay

Cultures were grown to a concentration of 1×10^7 cells/mL and the culture supernatant was syringe filtered with the (0.2 μm pore size) (Sartorius Stedim). SiHa (a cell line isolated from primary uterine tissue) cells were seeded in 96-well tissue culture plates containing Dulbecco's modified Eagle medium (DMEM) supplemented with 10% fetal bovine serum (FBS) at a density of 1×10^4 cells/mL. Following adherence, cells were treated with 10 μL of bacterial supernatant in 90 μL of DMEM medium with 10% FBS and cell survival was measured at 24, 48, and 72 h. After treatment, 1/10th volume of Cell Counting Kit-8 (CCK-8) was added to the cells, incubated for 4 h, and absorbance was measured at 450 nm using a plate reader (Varioskan™; Thermo Scientific, USA) (Prasad et al., 2020). Mitomycin (5 $\mu\text{g}/\mu\text{L}$) was used as positive control, while peptone water was used as vehicle control.

Transfection and T-cell factor (TCF) reporter assay

We tested for the antiproliferative activity of toxins A and B by polyethylenimine (PEI) transfection and T-cell factor (TCF) reporter assay. The reporter plasmids (TOPflash and FOPflash) were transfected to SiHa cells (6×10^5 cells) using PEI-mediated transfection. After 12 h of transfection, cells were treated with 200 μL of bacterial crude extracts corresponding to 10^6 CFU/mL and luciferase activities were measured after 36 h. SiHa cells co-transfected with pRL-TK (thymidine kinase basal promoter) were used as control (Roehl and Conrad, 1990). Dual luciferase reporter system (Promega) was used to quantify the Luciferase and Renilla by luminescence signal measurements (Lima et al., 2014).

Whole genome sequencing

Genomic DNA of CB MALS002, CD MALS003, and CT MALS001 was extracted using the standard phenol-chloroform protocol (Green and Sambrook, 2017) with some slight modifications. Cells were grown in anaerobic conditions, as mentioned earlier, for a period of 48 h and the cell pellets were lysed with 1.5 mL of pre-lytic buffer (20 mM Tris, 2 mM EDTA, 1% Triton X-100, and 60 $\mu\text{g}/\text{mL}$ of lysozyme) and incubated overnight at 37°C before genomic DNA was extracted. Quality and quantity of DNA were assessed with a Qubit 2.0 fluorometer (Thermo Fisher Scientific, USA).

Whole genome sequencing was performed on Ion Torrent Personal Genome Machine (PGM) (Thermo Fisher Scientific). One hundred nanograms of genomic DNA was fragmented to 400 bp using Ion Shear Plus Reagents followed by cleanup with Agencourt™ AMPure™ XP (Beckman Coulter, USA). Furthermore, the prepared libraries (100 pM) were amplified with Ion One Touch 2, enriched with Ion One Touch Enrichment System, and sequenced on an Ion Torrent Personal Genome Machine using a 318v2 chip following standard protocol (Utturkar et al., 2015).

Genome assembly, annotation, and comparison

After read quality check and trimming using FASTQC (Andrews, 2010) and FASTX toolkit (Gordon and Hannon, 2010), a reference-based assembly using SPAdes v3.13.0 (Prjibelski et al., 2020) was performed. References utilized for genome assembly are listed in Supplementary Table S1. Genome annotation was performed using National Center for Biotechnology Information (NCBI) Prokaryotic Genome Annotation Pipeline (PGAP, v6.1) (Tatusova et al., 2016). Assembled contig sequences were used as input for functional annotation using eggNOG-mapper (Huerta-Cepas et al., 2017) and Rapid Annotation using Subsystem Technology (RAST) (Aziz et al., 2008).

A maximum likelihood-based phylogenetic tree was constructed with bootstrap support of 1000 replicates using Molecular Evolutionary Genetics Analysis (MEGA) (Kumar et al., 2018). Nucleotide sequence level comparison between the clinical isolates and their respective reference genomes (Supplementary Table S1) was performed using Mauve (v20150226) (Darling et al., 2004). Annotation of raw sequences was performed using prokka v1.14.6 (Seemann, 2014) followed by pangenome analysis using roary v3.13.0 (Page et al., 2015) to obtain core and accessory genes. To predict single copy gene content, Benchmarking Universal Single-Copy Orthologue (BUSCO) v4.1.2 (Seppey et al., 2019) analysis was done using lineage-specific (*Clostridia* class level) dataset. Reference genomes used for BUSCO and pangenome analysis are listed in Supplementary Table S1.

Functional profiling

Genes coding for AMR was identified using AMRfinderPlus v3.10.5 (Feldgarden et al., 2019) and ABRicate v1.0.1 (Seemann, 2021). Unique hits across two approaches were identified and filtered ($\geq 75\%$ query cover and $\geq 70\%$ identity). Virulence genes were predicted with ABRicate tool compared to core and full data set of Virulence Factor Database (Chen et al., 2016) and visualized with CGView

(Stothard and Wishart, 2005). Genes corresponding to sporulation process were extracted from functional annotation (eggNOG-mapper) files manually. A blastp alignment was performed for common sporulation proteins between three strains to find sequence similarity.

Results

Biochemical and antimicrobial characterization

CB MALS002, CD MALS003, and CT MALS001 were revived and tested for their starch degradation and motility characteristics. *Clostridia* spp. have been shown to have differential starch degradation abilities. We observed saccharolytic activity (zones of clearance around bacterial growth) among CB MALS002 and CT MALS001 in decreasing order, respectively, while CD MALS003 did not display saccharolytic activity (Fig. 1a). CB MALS002, CD MALS003, and CT MALS001 displayed differential rates of swimming motility (Fig. 1b). In stab culture, the isolates showed varying degrees of motility, with CD MALS003 showing the highest swimming motility with vastly diffused growth followed by CB MALS002 and CT MALS001. In terms of antimicrobial profiles, CD MALS003 was resistant to vancomycin, while CB MALS002 and CT MALS001 were

sensitive. For ciprofloxacin, CB MALS002 was found to be resistant, while CD MALS003 and CT MALS001 were sensitive to the antibiotic (Supplementary Table S2).

Biofilm formation analysis

Biofilm formation in the three strains was measured with crystal violet staining. CD MALS003 showed enhanced biofilm production compared to CB MALS002 and CT MALS001. While dual cultures of CD MALS003 and CT MALS001 showed significantly higher levels of biofilm production compared to monocultures, CB MALS002 and CT MALS001 dual cultures showed similar levels of biofilm production as their respective individual cultures. Biofilm levels were higher in combined cultures, having the presence of all the three strains at 24 and 48 h postincubation (Fig. 2a).

Cytotoxic effect of *Clostridia* secretome on SiHa cells

Cytotoxic effect of culture supernatants of CB MALS002, CD MALS003, and CT MALS001 on SiHa cells was tested by CCK-8 assay. Different concentrations of the culture supernatant (crude secretome, 1:5 and 1:10 dilutions) were

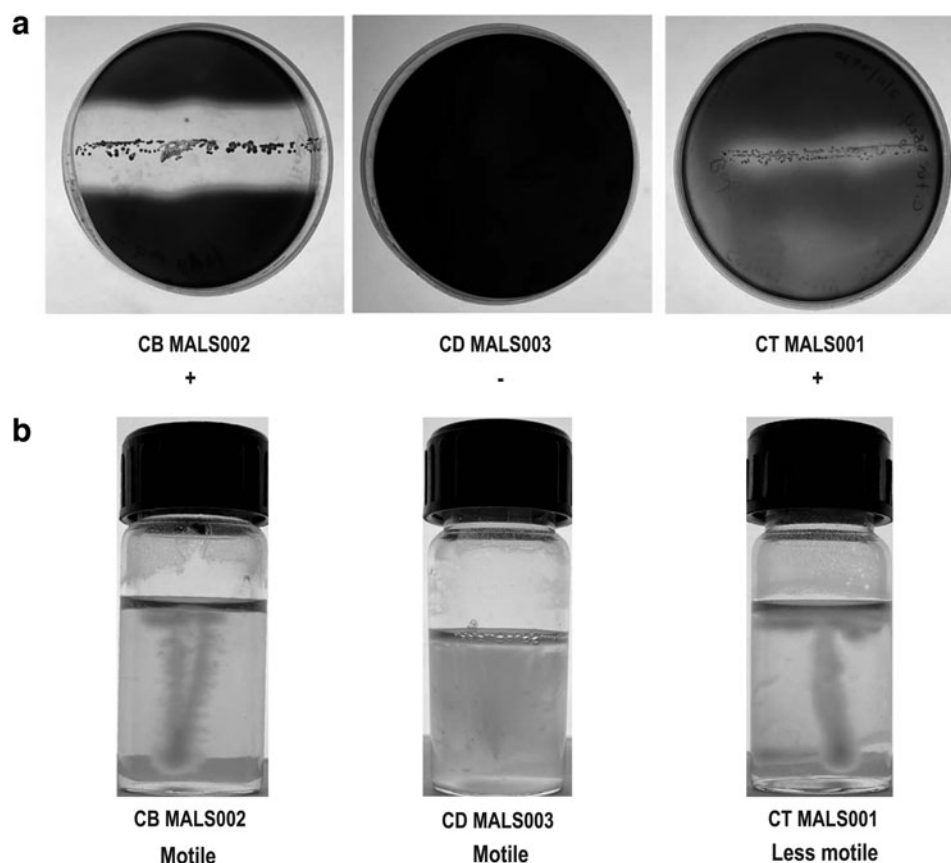


FIG. 1. Biochemical characterization. (a) Starch hydrolysis assay for CB MALS002, CD MALS003, and CT MALS001. Clear zones indicate ability of bacteria to hydrolyze starch. “±” indicates presence and absence of clear zones, respectively. (b) Motility test for CB MALS002, CD MALS003, and CT MALS001. Spreading phenotype was visually assessed to determine degree of motility. Diffuse spreading phenotype from the line of stab inoculum was considered visual confirmation of swimming motility. Experiments were performed in triplicates. CB, *Clostridium butyricum*; CD, *Clostridioides difficile*; CT, *Clostridium tertium*.

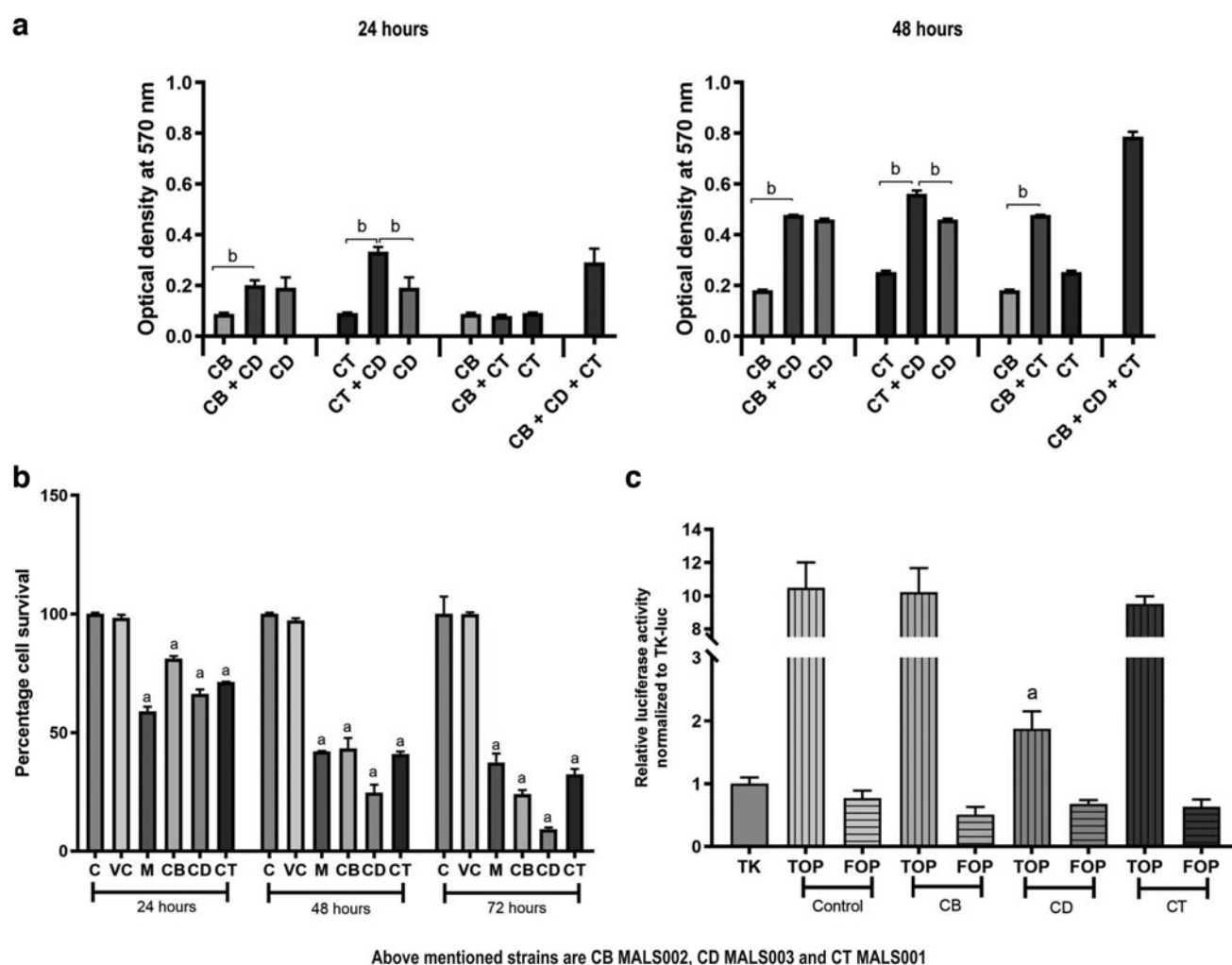


FIG. 2. Biofilm, cytotoxicity, and antiproliferative assay of CB MALS002, CD MALS003, and CT MALS001. (a) Detection of biofilm produced by three *Clostridia* in static condition using crystal violet assay at 24 and 48 h. CD MALS003 showed enhanced biofilm production compared to other strains. Combination of CB MALS002, CD MALS003, and CT MALS001 promoted biofilm formation compared to biofilm of monocultured and co-cultured species. Data presented as mean $\pm$ SE of $n=3$ independent experiment. Statistical significance was analyzed by student t test, $^b p \leq 0.01$. (b) Cytotoxicity of *Clostridia* crude supernatant on cell survival of SiHa cells was tested by CCK-8 assay at 24, 48, and 72 h. $^a p \leq 0.001$ compared to control. (c) Antiproliferative effect of *Clostridia* crude extracts on SiHa cells was tested by TCF reporter assay SiHa cells were co-transfected with either TOPFlash or FOPFlash luciferase reporter constructs to assess decrease/increase in proliferation of SiHa cells by changes in luciferase expression. Data presented as mean $\pm$ SE of $n=3$ independent experiment. Statistical significance was analyzed by student t test, $^a p \leq 0.001$. CCK-8, Cell Counting Kit-8; SE, standard error; SiHa, a cell line isolated from primary uterine tissue; TCF, T-cell factor.

tested. The crude secretome of all three bacterial strains showed significant cytotoxic effect at 24, 48, and 72 h (Fig. 2b), while the diluted extracts did not show significant difference in cytotoxicity compared to control (Supplementary Fig. S1).

Effects on Wnt pathway reporter gene expression by *Clostridia* toxins

The antiproliferative effect of CB MALS002, CD MALS003, and CT MALS001 secretome was investigated by β -catenin reporter assay using SiHa cells. The comparison of the ratio of TOPFlash to FOPFlash luciferase activity in SiHa cells treated with CD MALS003 secretome showed a significant downregulation in the β -catenin transcriptional reg-

ulatory response compared to control. However, there was no significant difference in expression in cells treated with CB MALS002 and CT MALS001 when compared to control (Fig. 2c).

Genome assembly, annotation, and whole genome comparison

Reference-based genome assembly and refining resulted in a genome size of 4.5 Mb for CB MALS002 with 125 contigs, 4.4 Mb for CD MALS003 with 132 contigs and 4.5 Mb for CT MALS001 with 169 contigs. Phylogenetic analysis of our sequences with related sequences confirmed that the strains belonged to CB, CD, and CT genomes (Supplementary Fig. S2). CT MALS001 genome harbored 4332

genes followed by CB MALS002 (4330) and CD MALS003 (4138). Sequence annotation identified 4186 proteins in CT MALS001, followed by CB MALS002 (4131) and CD MALS003 (3973). Strain CB MALS002 had the highest number of transfer RNA (83) sequences followed by CD MALS003 (77) and CT MALS001 (53). CB MALS002 also had the highest number of ribosomal RNA (23) sequences followed by CT MALS001 (7) and CD MALS003 (5) (Table 1).

The most abundant RAST SEED subsystem (functionally related protein families) across all *Clostridia* spp. belonged to “carbohydrates,” “amino acids and its derivatives,” and “protein metabolism” categories with an average of 16.85%, 15.57%, and 11.25%, respectively. Relative contribution of protein families involved in “dormancy and sporulation” was higher in CB MALS002 (4.1%) compared to CD MALS003 (2.1%, twofold decrease) and CT MALS001 (1.3%, threefold decrease).

A threefold increase of protein families involved in “phosphorus metabolism” and “nitrogen metabolism” was observed in CB MALS002 and CT MALS001 when compared with CD MALS003 (Supplementary Fig. S3a). Comparison of annotated genes using PGAP, RAST, and eggNOG-mapper showed that eggNOG could annotate higher number of genes (Supplementary Fig. S3b). A total of 757 protein-encoding genes were identified as “core proteome,” while a total of 167, 234 and 97 unique protein encoding genes, were found in CB MALS002, CD MALS003, and CT MALS001 genomes, respectively (Supplementary Fig. S3c). CD MALS003 shared 72.6% and 71.9% of genes with CB MALS002 and CT MALS001, respectively, while CB MALS002 and CT MALS001 shared higher number of protein-encoding genes.

Multiple genome alignment using Mauve highlighted several genomic rearrangements, including inversions, shifts, deletions, and gaps, in assembled genomes relative to reference genomes. CD MALS003 had an average of 99.67% identity with its reference genomes followed by CB MALS002 (99.35% identity), while the CT MALS001 genomes showed less identity (99.18%) (Supplementary Fig. S4).

Pangenome analysis

Pangenome analysis of three isolates with respective references showed 3228 core genes that were shared among five different CD genomes, while CB (2836 among 5 taxa)

TABLE 1. GENOME ASSEMBLY AND ANNOTATION FEATURES OF SEQUENCED STRAINS

Strain	<i>Clostridium</i> butyricum MALS002	<i>Clostridioides</i> difficile MALS003	<i>Clostridium</i> tertium MALS001
Total length (bp)	4,551,027	4,403,852	4,532,340
No. of contigs	125	132	169
N50 (bp)	107,911	187,823	70,613
GC content (%)	28.7	28.7	27.8
Genes	4330	4138	4332
Proteins	4131	3973	4186
rRNA	23	5	7
tRNA	83	77	53

rRNA, ribosomal RNA; tRNA, transfer RNA.

and CT (2741 among 4 taxa) genomes shared fewer core genes. Presence of accessory genes was highest among CT MALS001 genome with 1311 unique genes, out of which 884 genes were hypothetical, while 39 encoded putative proteins. CB MALS002 genome had 552 (331 hypothetical +18 putative) unique genes. CD reference genomes had fewer unique genes with strain FDAARGOS 267 having no unique gene, while ATCC 9689 strain had only one unique gene (Fig. 3a).

Among our strains, CD MALS003 had fewer unique genes in comparison with CB MALS002 and CT MALS001 (110, of which 92 were hypothetical). After removing hypothetical and putative genes from core genome, CD MALS003 assembly contained 1840 genes followed by CT MALS001 (1697 genes) and CB MALS002 (1594 genes). The core genome in individual strains ranged from 67–82% in CB genomes, 80–90% in CD genomes, and 65–77% in CT genomes (Fig. 3b).

Based on BUSCO analysis of 247 single copy genes from 385 *Clostridia* genomes, strains MALS001, MALS002, and MALS003 were found to have 244, 244, and 245 single copy genes, respectively (Fig. 3c). This included DHHA1 domain protein, which was absent in CD MALS003 and ATCC 9689 compared to the R20291 strain; pyridoxal phosphate homeostasis protein, which was absent in CB MALS002 compared to NBRC 13949 strain; and cytidine deaminase enzyme, which was absent in CT MALS001 when compared to Src5 strain. A total of 5 fragmented genes coding for helicase/exonuclease AddAB subunit *addA* in CD MALS003, 2,3-bisphosphoglycerate-independent phosphoglycerate mutase and Dephospho-CoA kinase in CB, and *recN* and phosphohydrolase in CT MALS001 were identified.

AMR profile

A total of 9 unique genes coding for AMR were predicted from all three strains. A variety of antibiotic resistance genes conferring resistance to glycopeptide, fluoroquinolone, streptogramin, β -lactam, and phenicol were recorded (Supplementary Table S3). Highest number of resistance genes were found in CD MALS003 (7), followed by CT MALS001 and CB MALS002, with one AMR gene each. No common AMR gene was found across all three strains.

AMR predicted for two antibiotic classes, fluoroquinolone and glycopeptide, were selected for phenotypic confirmation by minimum inhibitory concentration assay. Strain CB MALS002 showed phenotypic resistance to ciprofloxacin, despite no resistance gene being predicted for these drug classes (Supplementary Tables S2 and S3). Strain CD MALS003 showed phenotypic resistance to vancomycin. Concurrent with *in-silico* analysis five genes (*vanG*, *vanR*, *vanS*, *vanT*, and *vanZ1*) encoding resistance to vancomycin (Stogios and Savchenko, 2020) were predicted in CD MALS003 genome (Supplementary Tables S2 and S3).

Toxin/virulence profile

Sixty-six unique toxin or virulence genes were predicted across all three strains with *groEL*, *plr*, and *fbpA* being common among all three strains. Toxin genes were highest in CD MALS003 (54) followed by CT MALS001 (14) and CB MALS002 (13). In CD MALS003, a large cluster consisting of 31 toxin genes was identified, which coded for

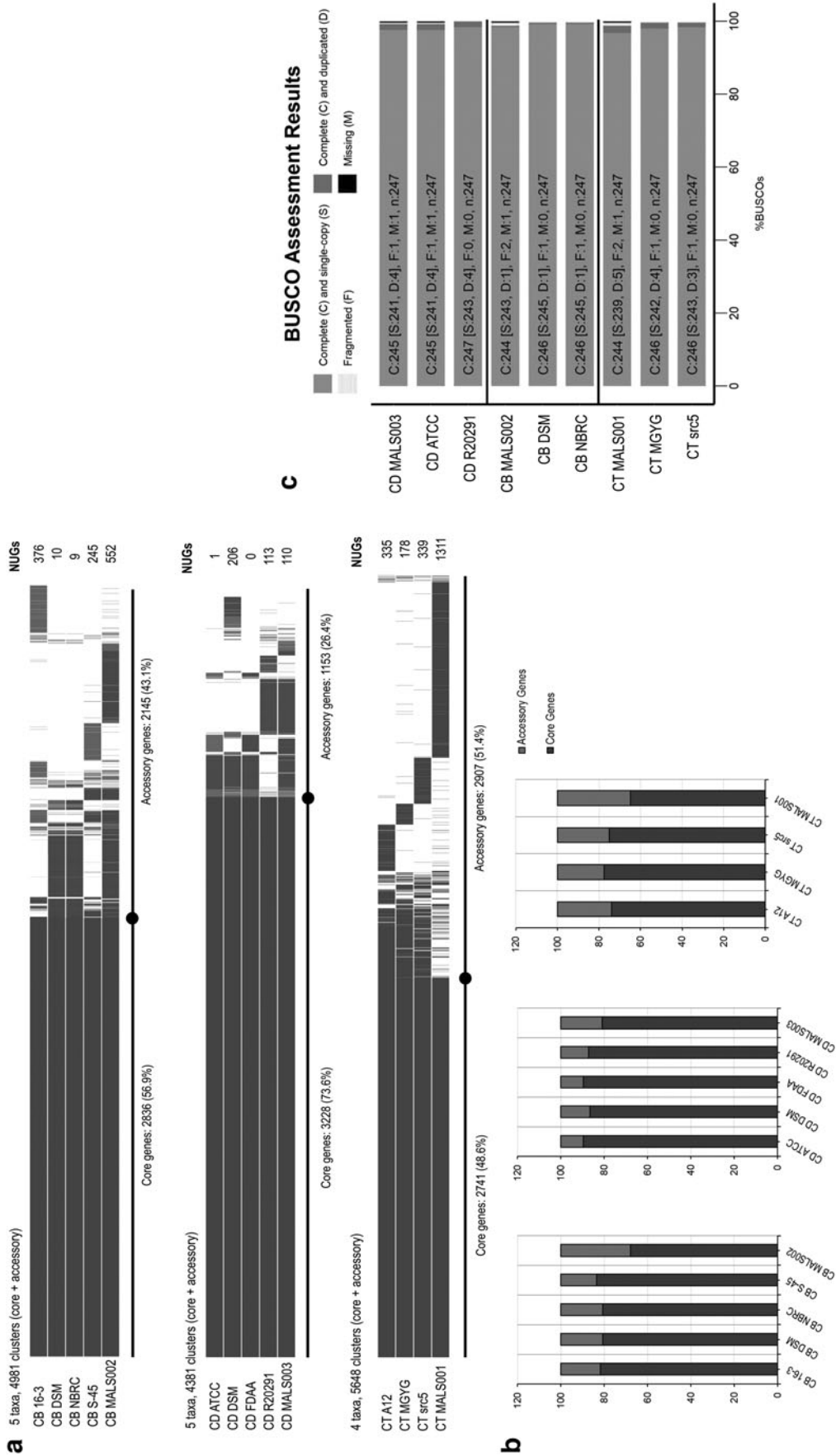


FIG. 3. Pangenome analysis of CB MALS002, CD MALS003, and CT MALS001. **(a, b)** Pangenome analysis for *Clostridia* genomes as performed with Roary. Parameters for selection: core genes: 99% $\leq$ strains $\leq$ 100% – accessory genes (15% $\leq$ strains $\leq$ 95%). **(c)** Genomic data completeness was analyzed with prediction of *Clostridia* single copy genes with BUSCO. Split of complete genes (single copy and duplicate), fragmented genes, and missing genes are shown in the bar graphs. BUSCO, Benchmarking Universal Single-Copy Orthologue; NUGs, number of unique genes.

flagellar proteins and few cell adhesion proteins (Supplementary Table S4). Two exotoxin-coding sequences located in pathogenicity locus (PaLoc) coding for toxin A (*tcdA*) and toxin B (*tcdB*), which are implicated in pathogenicity of CD, as well as three other accessory genes *tcdC*, *tcdE*, and *tcdR*, were present in our CD MALS003 strain.

Toxin genes unique to CB MALS002 were *Cbei_1707*, *Cbei_0023*, *pfoA*, *ureB*, *rmlA*, and *ZP_02950902*, while toxin genes unique to CT MALS001 were *nagK*, *pilT*, *CLL_A2400*, and *SSP0068*. Hyaluronidase exoenzyme (*nagK*) and hemolysin A (*CLL_A2400*) were major predicted toxins along with few flagellar proteins (*cheY*, *flhA*, *flp*, *fliS2*, *fliL*, and *pilT*) in CT MALS001 genome. In CB MALS002, perfringolysin O (*pfoA*), an exotoxin, and two hemolysins (*Cbei_1707* and *Cbei_0023*) were predicted along with very few flagellar proteins (*fliS1*, *fliL*, and *cheY*). Genomic locations of core and full toxins, AMR genes, and genes coding for flagellar assembly of all three strains were plotted (Fig. 4).

Sporulation genes

From functional annotation files obtained using eggNOG-mapper, a total of 105 genes corresponding to sporulation process were extracted (Supplementary Table S5). Following removal of duplicates based on gene copies in each isolate, 53 genes were retained, of which 26 genes were shared among all the three isolates (Supplementary Fig. S5a). Highest number of sporulation genes was present in CB MALS002 (48), followed by CT MALS001 (47) and CD MALS003 (32). A comparison of 26 common sporulation proteins using protein blast (blastp) identified *spoIIIE* to have the maximum sequence dissimilarity across the three strains, while RNA polymerase sporulation-specific sigma factors (*sigG*, *sigE*, *sigA*, and *sigK*) shared the highest sequence similarity (Supplementary Fig. S5b).

Discussion

The class *Clostridia* comprises heterogeneous species that are phylogenetically distantly related. The major human pathogens include CD, *Clostridium botulinum*, *Clostridium tetani*, and *Clostridium perfringens*, while several other species such as CT and CB are emerging as new pathogens and can cause human infections.

In this study, CB MALS002, CD MALS003, and CT MALS001 strains were characterized by culture-based and genomic methods. CB is a predominantly saccharolytic bacterium that plays a crucial role in the regulation of intestinal homeostasis and conversion of dietary fiber to beneficial short-chain fatty acids (SCFAs) (Zhang et al., 2018). Contrary to current literature, CD MALS003 from our study did not display any saccharolytic activity. However, CD has a diverse metabolic capacity and has been known to exhibit varying patterns of carbohydrate fermentation (Neumann-Schaal et al., 2019).

Biofilm production, which significantly contributes to virulence potential, was found to be higher in CD MALS003 strain compared to CB MALS002 and CT MALS001 (Fig. 2a). In addition, combined cultures of CT MALS001 and CD MALS003 produced significantly higher levels of biofilm when compared to their monocultures. However, biofilm levels of CD MALS003 on co-culture with CB MALS002 did not show any significant variation. While all

three strains displayed varied motility, CT MALS001 displayed limited motility in comparison to the diffuse phenotype observed in CB MALS002 and CD MALS003.

In addition to biofilm production and flagellar motility, which are critical for host colonization (Chen et al., 2019), AMR is a significant factor in determining pathogen epidemiology (Spigaglia et al., 2018). Our genomic data showed that CD MALS003 harbored five genes encoding vancomycin resistance (*vanG*, *vanR*, *vanS*, *vanT*, and *vanZ1*) (Stogios and Savchenko, 2020), while no such gene was detected in CT MALS001 and CB MALS002. This was concurrent with an observation of phenotypic resistance (Supplementary Table S2). However, despite possessing *cdeA* gene, which is known to encode resistance to ciprofloxacin and other fluoroquinolone drugs (Saxton et al., 2009), CD MALS003 displayed phenotypic susceptibility.

This phenotypic susceptibility, despite the presence of corresponding resistance genes, may be explained by the presence of compensatory mutations (Melnik et al., 2015) or by gene silencing by transcriptional control (Enne et al., 2006). However, these genes can get activated by metabolites produced by co-occurring bacteria such as CB and CT (Ferraris et al., 2012; Schmidt et al., 2015). Of note, genome analysis did not predict any resistance gene encoding fluoroquinolone resistance in CB MALS002 (Supplementary Table S3).

However, it demonstrated a high degree of phenotypic resistance toward the same. This phenotypic resistance in the absence of genotypic evidence in the form of presence of genes encoding AMR could hint toward the presence on “noninherited antibiotic resistance” (Levin and Rozen, 2006). This finding also emphasizes the need to explore emergence of AMR in relation with microbial communities (Bottery et al., 2020). CT MALS001 exhibited genomic and phenotypic susceptibility to both ciprofloxacin and vancomycin. This is concordant with the pathogenic profile of CT, which is largely considered a nonpathogenic strain that is occasionally found in clinical cases (Moore and Lacey, 2019; Steensma et al., 2011).

We found a total of 66 different toxin genes in the three isolates, with CD MALS003 harboring maximum number of toxin-coding genes (54), while CB MALS002 (13) and CT MALS001 (14) showed fewer toxin-coding genes. Based on genomic analysis, CD MALS003 was characterized as a toxigenic strain with the presence of *tcdA*, *tcdB* (encoding toxins A and B), *cdtA*, and *cdtB* (encoding binary toxin), while these toxin genes were absent in CB MALS002 and CT MALS001. Accordingly, CD MALS003 exhibited significant cytotoxicity *in vitro* compared to other tested strains. Studies indicate that CB strains can be used to prevent CDI as well as in improving gastric ulcers and other bacterial infections due to their ability to produce bacteriocins and SCFAs.

However, some toxigenic strains of CB have been implicated in botulism and necrotizing enterocolitis (Cassir et al., 2016a). In this study, even though CB MALS002 and CT MALS001 did not have any of the toxin genes commonly implied in the pathogenicity of CDI, we detected other toxin-coding genes coding for hemolysins such as *pfoA* (cholesterol-dependent cytolysin), which have been previously reported in *C. perfringens* (Verherstraeten et al., 2015). GroEL has been reported to play an important role in stimulating inflammatory response and is known to contribute

to bacterial infections (Ranford and Henderson, 2002). The presence of various genes coding for toxins and virulence factors along with reports on their ability to produce bacteriocins and butyric acid highlights the confusion in categorizing CB as a pathogen or a beneficial gut microbe (Cassir et al., 2016b).

CT, long considered a nontoxin producer, was occasionally found in clinical cases such as bacteremia and necrotizing fasciitis associated septicemia in patients with neutropenia and hematological malignancies (Shah et al., 2016), brain abscess (Lew et al., 1990), and meningitis (Kourtis et al., 1997), among others. However, recent reports indicate that CT isolated from stool samples of patients with diarrhea harbored sequences homologous to *tcdA*, *tcdB*, and other toxin-coding genes (Muñoz et al., 2019), suggesting their importance in disease outbreaks and pathogen emergence.

The presence of *nagK*, a bacterial hyaluronidase reported earlier from *C. perfringens* (Geier et al., 2021) in CT MALS001, demonstrates the extent of dynamic mobility among the clostridial genomes. Although bacterial hyaluronidase such as *nagH* has been reported in CT (Muñoz et al., 2019), to the best of our knowledge, *nagK* has only been reported from *C. perfringens* and CD, but not from CT strains (Low et al., 2018).

The importance of spore differentiation in pathogenesis of enteric pathogens cannot be underestimated. For instance, *spo0A* gene codes for a master regulator that turns on several downstream RNA polymerase sigma factors, including *sigG*, suggesting their role in persistence and transmission of highly resistant endospores, thereby providing the bacteria with high resistance and resilience to survive in the gut environment (Pereira et al., 2013). Interestingly the highest number of sporulation genes was present in CB MALS002 followed by CT MALS001 and CD MALS003.

In summary, surveillance of infectious diseases, even those with a predominant single pathogen such as CDI, may be severely limited if performed without considering microbial interaction with co-occurring pathogens such as CB and CT (Bottery et al., 2020; Ferraris et al., 2012). Several elements of microbial interactions such as extracellular vesicles and microbial volatiles have been known to influence gene expression of microbial partners leading to enhanced AMR, biofilm production, stress defense, immunomodulation, and pathogenic colonization, among several other roles that favor pathogen emergence and increase in disease severity (Schmidt et al., 2015; Tarashi et al., 2022).

Hence, it is critical to study the genomes of microbes known to co-occur with CD such as CB and CT to assess their genomic potential, so as to offer insights into preventing epidemic outbreaks and study the evolution of pathogens. Toward this, our study based on whole-genome sequence analysis of bacteria frequently associated with CDI (CD—known pathogen (Dawson et al., 2009), CB—known to be both beneficial and pathogenic (Cassir et al., 2016b), and CT—generally considered nontoxigenic (Moore and Lacey, 2019) was performed to identify potential virulence factors, AMR genes, mobile genetic elements, and sporulation factors, which can provide insights into emergence of pathogens with improved virulence, AMR, and overall pathogenic potential.

Such studies will further improve our understanding of the development of AMR, provide new avenues in genomic

monitoring of emerging pathogens, and offer better treatment strategies for crippling infectious diseases worsened by multidrug resistance such as CDI that threatens planetary health.

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Authors' Contributions

Conceptualization: K.S., M.B., and T.S.M.; investigation: P.S.; software: A.S.T.; validation and visualization: P.S. and A.S.T.; writing—original draft: T.S.M. and A.J.; writing—review and editing: T.S.M., A.B., M.B., and K.S.; and supervision: T.S.M., A.B., M.B., and K.S.

Data Availability

Genomes of *C. butyricum* strain MALS002 (CB), *C. difficile* strain MALS003 (CD), and *C. tertium* strain MALS001 (CT) are submitted to NCBI genome database under BioProject ID PRJNA820142 and with following Genome accession numbers: JALGRX000000000 (CB), JALGRY000000000 (CD), and JALGRW000000000 (CT).

Author Disclosure Statement

The authors declare they have no conflicting financial interests.

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Supplementary Material

Supplementary Figure S1
Supplementary Figure S2
Supplementary Figure S3
Supplementary Figure S4
Supplementary Figure S5
Supplementary Table S1
Supplementary Table S2
Supplementary Table S3
Supplementary Table S4
Supplementary Table S5

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Address correspondence to:
 Thokur Sreepathy Murali, PhD
 Department of Biotechnology
 Manipal School of Life Sciences
 Manipal Academy of Higher Education
 Manipal, Karnataka 576104
 India

E-mail: murali.ts@manipal.edu

Abbreviations Used

AMR = antimicrobial resistance
BHIA = Brain Heart Infusion agar
CB = *Clostridium butyricum*
CCK-8 = Cell Counting Kit-8
CD = *Clostridioides difficile*
CDI = *C. difficile* infections
CLSI = Clinical and Laboratory Standards Institute
CT = *Clostridium tertium*
DMEM = Dulbecco's modified Eagle medium
EUCAST = European Committee on Antimicrobial
Susceptibility Testing

FBS = fetal bovine serum
MIC = minimal inhibitory concentration
NUGs = number of unique genes
PEI = polyethylenimine
PGAP = Prokaryotic Genome Annotation Pipeline
RAST = Rapid Annotation using Subsystem
Technology
rRNA = ribosomal RNA
SCFAs = short-chain fatty acids
SE = standard error
tRNA = transfer RNA
VFDB = Virulence Factor Database